

Health-related outcomes of chronic disease management program for diabetes mellitus in the Indonesian Eastern border region

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ABSTRACT

BACKGROUND The Indonesian chronic disease management program (PROLANIS) was introduced to provide structured chronic disease management for hypertension and type 2 diabetes (T2D) at the primary healthcare level. This study aimed to evaluate the health-related outcomes of PROLANIS among T2D patients in the eastern border region.

METHODS This retrospective cohort study used secondary data from PROLANIS reports of all primary healthcare facilities in Atambua Regency, East Nusa Tenggara Province, Indonesia. Inclusion criteria were all T2D patients who participated in routine 6-month evaluations conducted between June 2023 and December 2023. The assessed parameters and their defined targets were body mass index (BMI) (18.5–23 kg/m²), systolic blood pressure (SBP) (<140 mmHg), diastolic blood pressure (DBP) (<90 mmHg), hemoglobin A1c (HbA1c) (<7%), total cholesterol (TC) (<200 mg/dl), high-density lipoprotein (HDL) (>40 mg/dl for males and >50 mg/dl for females), low-density lipoprotein (LDL) (<100 mg/dl), and triglycerides (TGs) (<150 mg/dl). Wilcoxon signed-rank and paired t-tests were used for statistical analyses.

RESULTS There were 51 T2D patients included in the analyses. Of these, 84% achieved SBP and 96% achieved DBP targets. However, only 45% achieved HbA1c, and 33% achieved BMI targets. For lipid profiles, 55%, 51%, 47%, and 41% achieved TC, LDL, HDL, and TG targets, respectively. No significant differences were found between baseline and follow-up across these parameters (all $p > 0.05$).

CONCLUSIONS In the border region, while PROLANIS maintains blood pressure levels, it does not significantly improve other health-related outcomes in T2D patients. This underscores the need to strengthen program intensity, support adherence, and address contextual barriers to achieve meaningful metabolic improvements.

KEYWORDS chronic disease, government programs, health services, Indonesia, primary healthcare, type 2 diabetes mellitus

Chronic noncommunicable diseases, particularly type 2 diabetes (T2D), are becoming one of the most rapidly increasing global health crises of the 21st century.¹ T2D, which is characterized by hyperglycemia, is predicted to continue to increase in the coming

years, especially in low- and middle-income countries. According to data from the International Diabetes Federation in 2021, approximately 10.5% of adults aged 20–79 years worldwide suffer from diabetes. This number is projected to increase by 12.2% by 2045.²

In Indonesia, the National Health Survey has reported that the prevalence of diabetes has surged over the past decade, increasing from 5.7% in 2007 to 11.7% in 2023, placing immense pressure on the national healthcare system.^{3,4} Addressing this growing burden requires innovative approaches that go beyond conventional methods, since these methods disproportionately affect underserved populations in Indonesia. East Nusa Tenggara, a part of Indonesia's eastern border, reports limited access to healthcare services for chronic diseases, despite having prevalence rates comparable to the national average, owing to its geographic isolation and disparities in infrastructure.

Recognizing the increasing burden of chronic diseases, the Indonesian government introduced the chronic disease management program, *Program Pengelolaan Penyakit Kronis* [PROLANIS], in 2010. PROLANIS follows an integrated care model that combines routine medical consultations with peer support groups, lifestyle education, and continuous monitoring. Clinical evaluations were performed every 6 months to assess parameters such as blood glucose, blood pressure (BP), lipid profiles, and kidney function.³ Additionally, the evaluation reinforce education on lifestyle modification and symptom management while providing patients with the opportunity to discuss health goals, challenges, and adjustments to their care plans. This coordinated approach was designed to empower patients to play a more proactive role in managing their condition and improving long-term health outcomes. While this approach has shown promise, its success is dependent on factors such as patient adherence, engagement in educational programs, and regular attendance at routine clinical evaluations.^{5,6} However, evidence on the health-related outcomes of PROLANIS remains limited, with some reports showing that it improves health-related outcomes, whereas others show that it does not.⁷

The highest coverage of healthcare for chronic diseases is concentrated in the Java region, whereas the rural border regions of Indonesia have a limited healthcare infrastructure, making the implementation of programs such as PROLANIS more challenging. Although these regions are challenged by geographic remoteness and socioeconomic inequality, they offer opportunities for impactful community-based health interventions.^{6,8,9} Moreover, little is known about the clinical outcomes of PROLANIS in border regions because most studies have focused on urban

populations with better healthcare access. Rural border settings differ substantially because they face longer travel distances to primary care, fewer health workers, and limited diagnostic and medication availability, all of which reduce program participation and continuity of care compared to urban areas. This study addresses this gap by evaluating the clinical outcomes of PROLANIS among patients with T2D in the eastern rural border region, as health-related outcomes reflect the program's effectiveness. The findings of this study may provide insights into similar underserved settings.

METHODS

Study design and population

This retrospective observational cohort study was carried out in Atambua Regency, a rural area near the Indonesian border in East Nusa Tenggara Province, Indonesia, where access to diabetes care is limited, and PROLANIS serves as the main structured chronic disease management program. This study used secondary data from all the PROLANIS group reports in the Atambua Regency. All patients with T2D aged ≥ 18 years enrolled in PROLANIS who attended the routine 6-month evaluation in June 2023 were included. Those who discontinued membership, died, or did not attend the 6-month follow-up evaluation in December 2023 were excluded. Follow-ups were operationalized through attendance reports from the PROLANIS group reports.

As all eligible participants were included in the total sample, a sample size calculation was not performed prior to data collection. The final sample consisted of 53 participants because only this subgroup had completed paired clinical data from both the June 2023 baseline assessment and December 2023 follow-up visit. This number also reflects the limited overall PROLANIS enrollment and the high attrition characteristics of rural Atambua, thereby underscoring that the included subjects represent the active and continuously engaged segment of the T2D population in the Atambua Regency.

Evaluated parameters

The evaluated parameters and their defined targets were body mass index (BMI) (18.5–23 kg/m²), systolic blood pressure (SBP) (<140 mmHg), diastolic blood pressure (DBP) (<90 mmHg), hemoglobin

A1c (HbA1c) (<7%), total cholesterol (TC) (<200 mg/dl), high-density lipoprotein (HDL) (>40 mg/dl for males and >50 mg/dl for females), low-density lipoprotein (LDL) (<100 mg/dl), and triglycerides (TGs) (<150 mg/dl). T2D diagnoses were established by primary care physicians according to Indonesian Society of Endocrinology (PERKENI) and American Diabetes Association criteria, using either (i) fasting plasma glucose ≥ 126 mg/dl, (ii) 2-hour post-oral glucose tolerance test plasma glucose ≥ 200 mg/dl, (iii) HbA1c $\geq 6.5\%$, or (iv) random plasma glucose ≥ 200 mg/dl with symptoms. The defined targets and criteria were based on the 2021 guidelines of the PERKENI.¹⁰ These parameters were obtained from routine 6-month evaluations. Next to the parameters mentioned above, data for serum creatinine that reflects kidney function were also obtained and converted to estimated glomerular filtration rate (eGFR) (ml/min/1.73 m²) using the Chronic Kidney Disease Epidemiology Collaboration 2009 equation.¹¹ The primary outcome was continuous changes in the evaluated parameters. The secondary outcome was a change in the proportion of participants who achieved the defined target for each parameter.

To ensure standardization and comparability across facilities and over time, all measurements followed routine national primary care procedures. BP was measured in a seated position after at least 5 min of rest using automated sphygmomanometers available at each facility. The BMI was assessed using calibrated digital scales and stadiometers. BMI and BP were measured by healthcare workers at local primary care facilities, and all laboratory parameters were analyzed by designated local third-party laboratories. Other details such as age, sex, and baseline characteristics were obtained from the PROLANIS report.

Statistical analysis

Statistical analysis was performed using R software version 4.3.2 (R Foundation for Statistical Computing, Austria). The normality of the data was assessed using the Shapiro–Wilk test. Normally distributed variables were expressed as means (standard deviation), non-normally distributed variables as medians (interquartile range), and categorical variables as frequencies (valid percentages). Changes in the evaluated parameters were analyzed using paired t-tests or Wilcoxon signed-rank tests depending on the data distribution. Statistical significance was assessed using a two-sided $\alpha = 0.05$, and 95% confidence intervals were reported where applicable. Participants lost to follow-up were excluded from the analyses, and only those with paired baseline and follow-up measurements were included. No data were missing for the included participants. Binary target variables were generated by applying predefined thresholds from the 2021 PERKENI guidelines to each evaluated parameter.¹⁰ Outliers were reviewed through visual inspection of the distribution plots and verified against the original source records. All values were considered clinically plausible, and no exclusion was deemed necessary. No subgroup or interaction analyses were prespecified or conducted because of the limited statistical power. This study adhered to the principles of the Declaration of Helsinki and was approved by the Health Research Ethics Committee of the Faculty of Medicine, Universitas Airlangga (No: 298/EC/KEPK/FKUA/2023), and the Atambua Regency government (No: DPMPSTSP.5304/038/SKP/XI/2023) prior to data collection.

Bias

To reduce selection bias, a total sampling strategy was used, whereby all eligible PROLANIS participants

Figure 1. Flow diagram of participant selection and 6-month follow-up among adults with T2DM enrolled in the PROLANIS program of Atambua, Indonesia (June–December 2023). PROLANIS=Program Pengelolaan Penyakit Kronis; T2DM=type 2 diabetes mellitus

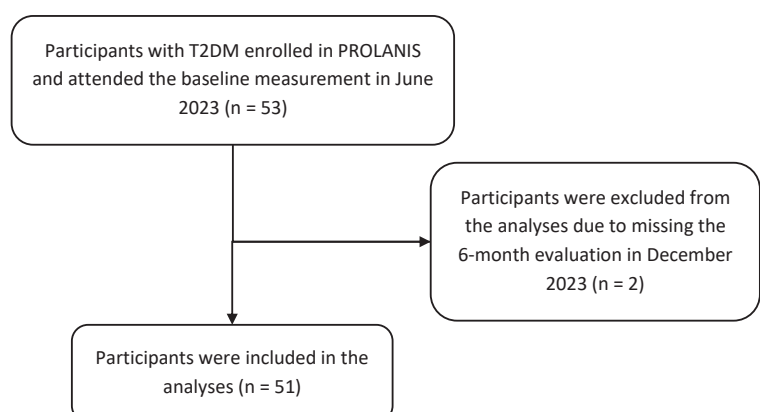


Table 1. Baseline characteristics of T2D patients enrolled in the chronic disease management program (PROLANIS) across primary-care facilities in Atambua, Indonesia

Baseline characteristics	N = 51
Age (years), mean (SD)	62 (11)
Male sex, n (%)	23 (45)
Education level up to senior high school, n (%)	30 (59)
History of hypertension, n (%)	9 (18)
Duration of T2D <5 years, n (%)	51 (100)
Monthly attendance, n (%)	
2x/month	24 (47)
3x/month	21 (41)
4x/month	6 (12)
Pharmacological therapy for T2D, n (%)	
1 OAD	22 (43)
2 OADs	22 (43)
Insulin	5 (10)
Combination of OAD and insulin	2 (4)
Clinical parameters, median (IQR)	
BMI (kg/m ²)	23.8 (22.36–26.30)
SBP (mmHg)	130 (113–140)
DBP (mmHg)	80 (70–81)
TC (mg/dl)	197 (182–224.5)
LDL (mg/dl)	100 (97–119)
HDL (mg/dl)	42 (38–49.5)
TG (mg/dl)	159 (138.5–184.5)
HbA1c (%)	7.3 (6.30–9.75)
eGFR (ml/min/1.73m ²)	86.23 (63.12–115.83)
Prevalence of patients with optimal metabolic control, n (%)	
BMI between 18.5–23 kg/m ²	17 (33)
SBP <140 mmHg	43 (84)
DBP <90 mmHg	49 (96)
TC <200 mg/dl	28 (55)
LDL <100 mg/dl	26 (51)
HDL >40 mg/dl (male) or >50 mg/dl (female)	24 (47)
TG <150 mg/dl	21 (41)
HbA1c <7%	23 (45)
eGFR >45 ml/min/1.73m ²	47 (92)

BMI=body mass index; DBP=diastolic blood pressure; eGFR=estimated glomerular filtration rate; HbA1c=hemoglobin A1c; HDL=high-density lipoprotein; IQR=interquartile range; LDL=low-density lipoprotein; OAD=oral antidiabetic drug; PROLANIS=Program Pengelolaan Penyakit Kronis; SBP=systolic blood pressure; SD=standard deviation; T2D=type 2 diabetes; TC=total cholesterol; TG=triglyceride. Optimal metabolic control was based on the 2021 guidelines of the Indonesian Society of Endocrinology (PERKENI). Data distribution for the continuous variables was assessed using the Shapiro–Wilk test

who attended the June 2023 evaluation were included. Measurement bias was minimized by confirming that measurements using standardized assessment procedures, calibrated equipment, and laboratory analyses were performed by accredited regional laboratories following consistent procedures at both time points.

RESULTS

Baseline characteristics

A schematic representation of the patient selection is shown in Figure 1. Of 53 patients enrolled in PROLANIS, 2 missed the 6-month evaluation and no deaths/discontinuations were recorded, leaving 51 patients for the analyses. Details of the baseline characteristics of the patients are summarized in Table 1.

Evaluated parameters

Most patients displayed favorable BP control, with 84% maintaining an SBP of <140 mmHg and 96% maintaining a DBP of <90 mmHg. In terms of lipid profiles, 55% of the patients had TC levels within the recommended limits and 51% had LDL cholesterol levels under control. Furthermore, only 47% achieved the target HDL levels and 41% had TG levels within the desired range. Glycemic control, indicated by HbA1c levels below 7%, was achieved in 45% of patients. Kidney function appeared to be well preserved among the patients, with 92.16% showing an eGFR of 45 ml/min/1.73 m². Additionally, 33% of the patients maintained their BMI within the optimal range (Table 1).

Follow-up outcomes

Changes in the evaluated clinical parameters indicated stability across most variables (Figure 2). There were no significant changes in clinical parameters between baseline and follow-up. The median levels of SBP (130 mmHg, $p = 0.530$), DBP (80 mmHg, $p = 0.300$), HDL (42 mg/dl, $p = 0.962$), and HbA1c (7.3%, $p = 0.425$) between baseline and follow-up remained unchanged. The median BMI (23.8 to 25.15 kg/m², $p = 0.448$), TG (159 to 167 mg/dl, $p = 0.435$), and TC (197 to 199 mg/dl, $p = 0.854$) levels showed a slight but non-significant increase, whereas median LDL (100–98 mg/dl, $p = 0.152$) and eGFR levels (86.2 to 83.6 ml/min/1.73 m², $p = 0.372$) showed a slight but non-significant decrease. The percentage of patients who achieved optimal

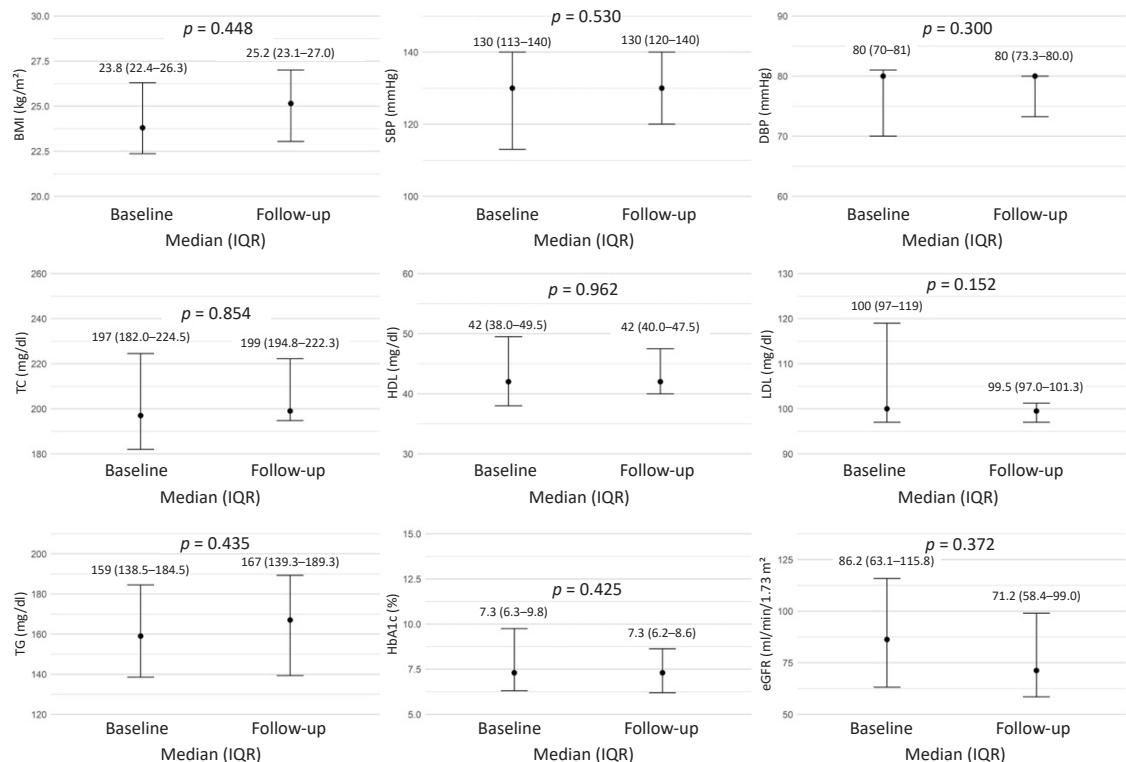


Figure 2. Baseline (June 2023) vs. 6-month (December 2023) changes in clinical markers among adults with T2D enrolled in PROLANIS, Atambua, Indonesia. Two-sided Wilcoxon signed-rank tests were used for the statistical analyses for all variables, as all variables were skewed. A $p < 0.05$ was considered statistically significant. BMI=body mass index; DBP=diastolic blood pressure; eGFR=estimated glomerular filtration rate; HbA1c=hemoglobin A1c; HDL=high-density lipoprotein; IQR=interquartile range; LDL=low-density lipoprotein; SBP=systolic blood pressure; T2D=type 2 diabetes; TC=total cholesterol; TG=triglyceride

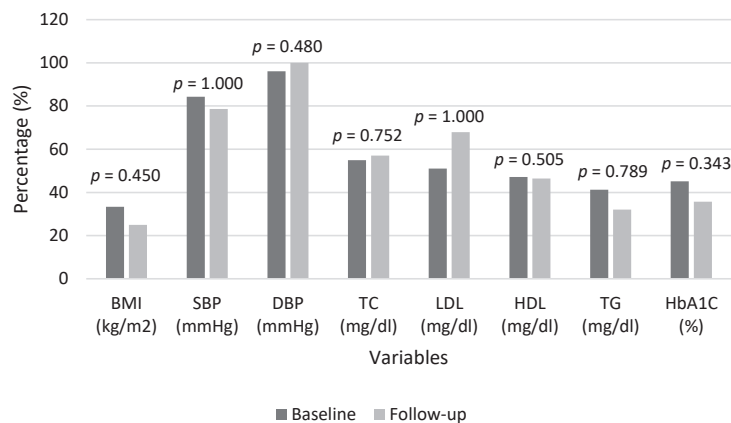


Figure 3. Trends in the percentage of participants who achieved optimal metabolic control at baseline and during the follow-up assessment. McNemar tests were used for the statistical analyses for all variables. Optimal metabolic control was based on the 2021 guidelines from the Indonesian Society of Endocrinology (PERKENI). BMI=body mass index; DBP=diastolic blood pressure; HbA1c=hemoglobin A1c; HDL=high-density lipoprotein; LDL=low-density lipoprotein; SBP=systolic blood pressure; TC=total cholesterol; TG=triglyceride

metabolic control during the baseline and follow-up periods is presented in Figure 3.

The results showed that a high proportion of patients achieved BP targets, whereas the proportion of patients achieving glycemic and lipid profile targets were more variable. These findings suggest that the

clinical outcomes varied across metabolic parameters during the study period. Notably, these outcomes were derived from routinely collected monitoring data within PROLANIS, which primarily captures clinical indicators. Although the dataset included clinical parameters of patients, information on program

implementation processes such as health education activities, frequency of sessions, and adherence tracking was not documented.

DISCUSSION

These findings highlight the stability of clinical profiles among patients enrolled in PROLANIS over time. Most patients achieved good BP control. However, only 45% of patients attained glycemic control (HbA1c <7%), and approximately half of them met lipid control targets. These outcomes suggest that while PROLANIS effectively manages BP, significant gaps remain in glycemic and lipid control. Excess BMI was also common among patients, consistent with other studies indicating an increase in the prevalence of overweight individuals within rural populations in Indonesia.¹² Importantly, when linked back to the objective of assessing 6-month outcomes, the lack of measurable improvement indicates that clinical parameters remained relatively stable over the follow-up period. This stability may be influenced by factors such as program intensity, varying levels of participant adherence, and relatively short duration of observation, all of which can limit the detection of meaningful changes in glycemic or lipid profiles. Taken together, these considerations provide context for interpreting the “no improvement” pattern and suggest that longer follow-up or more intensive program components are needed to demonstrate measurable metabolic changes.

In terms of HbA1c control, only a limited number of participants achieved levels of <7%. This indicates that T2D control remains a challenge in rural areas, a finding supported by similar research showing high HbA1c values, particularly among the elderly in rural populations.¹³ Similarly, the relatively low proportion with normal lipid profiles echoes data showing a high prevalence of dyslipidemia across ethnic groups in Indonesia (up to 36%).¹⁴ In contrast, kidney function was generally preserved, most participants maintained good eGFR values, likely because the majority had been diagnosed with T2D for less than 5 years, in line with evidence that significant kidney decline typically occurs after >10 years of disease.¹⁵ These mixed outcomes indicate the program’s strengths in maintaining clinical stability, and underscore ongoing challenges in achieving comprehensive metabolic control.

PROLANIS, introduced by the Indonesian government, is designed to manage chronic diseases, such as T2D, in primary healthcare settings. Although the model has been effective in urban and semi-urban areas, its implementation in rural border regions remains constrained by contextual challenges. One of the major obstacles is geographical disparity, as the eastern border areas of Indonesia are characterized by difficult terrain, remote islands, and a scattered population.⁷ Patients with T2D in these areas often need to travel long distances to reach the nearest healthcare center, which complicates their ability to participate in regular health check-ups, receive medication, and attend health education sessions offered by PROLANIS. Consequently, adherence to diabetes care protocols remains low, increasing the risk of complications such as retinopathy, nephropathy, and cardiovascular diseases.

In addition to geographical challenges, the lack of healthcare infrastructure exacerbates the difficulty of managing chronic conditions such as diabetes in border regions. Many healthcare facilities in these areas are under-resourced, with a shortage of medical equipment, such as blood glucose monitoring equipment, limited availability of healthcare workers, and inadequate diagnostic capabilities.^{16–18} As a result, even when patients with diabetes can access healthcare centers, the quality of care they receive may be compromised. Additionally, the limited availability of insulin, and other essential supplies further exacerbates difficulties in effectively managing diabetes.

Socioeconomic disparities in rural border regions also pose significant barriers to the successful implementation of PROLANIS. The population in these areas often experiences high levels of poverty, low levels of education, and poor health literacy, which hinder effective participation in chronic disease management programs.^{5,6,8} Many patients with diabetes cannot afford transportation to healthcare facilities or the necessary medications and dietary modifications to manage their condition effectively. Additionally, a low health literacy indicates that patients may not fully understand the importance of regular monitoring and adherence to treatment protocols, leading to suboptimal disease management. Furthermore, cultural beliefs and practices often influence patient engagement in diabetes management programs. In some rural communities, traditional remedies and healers are preferred over modern

medical treatments.^{19,20} This reliance on traditional medicine may hinder the acceptance of programs, such as PROLANIS, highlighting the need for culturally sensitive approaches that integrate traditional beliefs with modern healthcare practices. Additionally, many primary care providers lack incentives to actively recruit patients, which reduces the program's reach and impact. These structural and socioeconomic constraints explain why the outcomes observed in this rural border setting differ from those in urban and semi-urban areas, where better healthcare access, higher health literacy, and stronger program infrastructure typically support more consistent glycemic management and program participation.

Nonetheless, in rural border areas with limited healthcare access, the structured monitoring and prevention of metabolic parameters using PROLANIS may have a significant impact. Early detection and timely management of diabetes-related complications can reduce hospital admissions, lower the risk of severe outcomes, and improve quality of life.²¹ Findings from several rural regions suggest that routine monitoring and timely interventions through the PROLANIS have contributed to better BP and lipid control, potentially lowering the development of diabetes-related complications.^{6,8,22,23}

Obesity is a known risk factor for T2D, and weight management is a key component of PROLANIS. This program encourages physical activity and promotes dietary changes aimed at reducing calorie intake and improving nutrition. Community-based physical activities such as group exercise sessions organized by local health workers can enhance weight management. In rural border areas, where access to healthy food may be limited, these interventions are crucial for helping patients manage their weight and reduce obesity-related complications. However, the program lacks sufficient infrastructure and incentives to effectively implement these interventions in remote regions, where access to both healthy foods and regular physical activity is often limited.^{22,24,25} Given the multiple outcomes assessed and the short follow-up period, these findings should be interpreted primarily as program monitoring rather than as a measure of program effectiveness. Notably, the observed pattern of improvements, showing relatively better BP control than glycemic control, aligns with reports from urban settings, suggesting a broader trend across PROLANIS implementation.

By assessing the effectiveness of PROLANIS in this context, this study provides valuable insights into chronic disease management, particularly regarding how chronic disease programs function in geographically and socioeconomically disadvantaged communities. These findings provide insights into policy adaptations to suit underserved populations better. Regarding generalizability, the findings of this study are most applicable to rural or border primary care settings with program structures comparable to those of PROLANIS. Caution is warranted when extending these results to provinces with different referral pathways, supply chain conditions, or broader health system characteristics. However, its limitations include its observational design and the lack of a control group, which restricts the attribution of observed outcomes directly to PROLANIS participation. The small sample size and short follow-up duration also limit the generalizability of these results to other populations and contexts. Additionally, this study did not evaluate qualitative aspects, such as patient satisfaction or barriers to program adherence, which could provide a more comprehensive understanding of program effectiveness. Future studies should incorporate controlled designs and qualitative assessments to address these issues.

In summary, the implementation of PROLANIS in the eastern rural border region of Indonesia has helped maintain BP levels among patients with T2D. However, challenges remain in achieving optimal glycemic and lipid profile control, weight management, and consistent attendance. Limited healthcare infrastructure and access along with socioeconomic barriers continue to impede comprehensive diabetes management in this region. Addressing these issues through targeted interventions, including improving health literacy, expanding healthcare resources, and strengthening infrastructure, is essential. Multisectoral collaboration and technology integration could significantly extend the reach and impact of PROLANIS, thereby enhancing the quality of care. These efforts can contribute not only to improved health outcomes in the rural border region of Indonesia, but also serve as a model for resource-limited countries facing similar public health challenges.

Conflict of Interest

The authors affirm no conflict of interest in this study.

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